Q.28 Write any five properties of an Ideal Refrigerant.

Q.29 Write a short note on window air-conditioner.

Q.30 Write various advantages of Solar Power refrigeration system over vapour Compression refrigeration system.

Q.31 Explain the process of sensible heating in psychrometry.

Q.32 What is the effect of Sub cooling the liquid on the performance of a Vapour Compression system?

Q.33 Define evaporator and name the various types of evaporators.

Q.34 Explain reversed Carnot cycle refrigerator.

Q.35 Explain the working of vapour compression refrigeration system with neat sketch.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the working of Split Air-Conditioning system with the help of neat sketch. Give its advantages.

Q.37 Explain Solar power refrigeration system with the help of neat sketch.

Q.38 Explain the working of reciprocating Compressor with the help of neat sketch.

Roll No. ....

5th Sem / Mech. Engg. (MSIL)
Subject:- Refrigeration and Air Conditioning

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Solid carbon dioxide is known as

a) Moist air b) Refrigeration effect

c) Dry Ice d) Refrigeration

Q.2 The Cooling effect produced by a refrigeration system in a given time is called

a) C.O.P b) Refrigeration effect

c) Air Conditioning d) TR

Q.3 Air refrigerator works on

a) Carnot cycle

b) Reversed Carnot cycle

c) Bell Coleman cycle

d) Both (b) and (c)

Q.4 The chemical formula of ammonia is

a) NH_3 b) CO_2

c) NH d) CH_4

Q.5 The refrigerant number of Carbon dioxide is

a) R-717 b) R-22

c) R-744 d) R-12

Q.6 The natural convection air-cooled condensers are used in

a) Water coolers

b) Refrigerators

c) Room air-Conditioner

d) All of the above

Q.7 The evaporator generally used in home freezer, ice cream Cabinets etc. is

a) Plate evaporator

b) Finned evaporator

c) Shell and tube evaporator

d) Shell and coil evaporator

Q.8 During sensible heating of air, the humidity ratio

a) Increase b) Constant

c) Decrease d) None of the above

Q.9 The commonly used refrigerant in window air-conditioner is

a) R-12 b) R-13

c) R-717 d) R-22

Q.10 Air without moisture Content is called

a) Moist air b) Dry air

c) Humidity d) Saturated air

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Define one ton of Refrigeration system.

Q.12 Define DPT.

Q.13 Name some Secondary refrigerants.

Q.14 Write the functions of expansion valve.

Q.15 What is evaporator.

Q.16 Define Moist air.

Q.17 Define Humidity.

Q.18 Define Sensible Cooling.

Q.19 What is the refrigerant number of Carbondioxide.

Q.20 Define C.O.P of a refrigeration system.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the function of overload protector.

Q.22 Draw the P-h diagram of Simple Vapour Compression refrigeration cycle.

Q.23 Write the various properties of R-744.

Q.24 Explain psychrometry chart.

Q.25 Explain domestic electrolux refrigeration system with neat sketch.

Q.26 Write the function of condenser. Explain any one.

Q.27 Explain centrifugal compressor.